# The Instrument Is Inside:
# Reflexivity as a Methodological Constraint

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September 2026

**Abstract**

Process accounts of nature—that what persists are patterns maintained by activity rather than substances that happen to change—are old and, taken alone, hard to test. This paper argues that such accounts have one consequence which is not merely interpretive: if a system acts on a world that contains it, then any instrument built to study that system is also inside it, and this predicts a specific, recurring pathology in measurement.

The prediction is that quantities fixed by the *construction* of an experiment will be read as properties of the *system*, and that this will happen preferentially when the quantity agrees with the hypothesis under test. We document seven instances from our own laboratory—an item-balance floor read as subject incapacity, an unreachable code branch read as an inert monitor, a randomised trigger reported as a diagnosis rate, and three others. Every one agreed with what we expected. None was audited until an external control forced it.

We argue this is not a catalogue of carelessness but the expected signature of reflexive measurement, that it identifies a class of controls which are cheap and rarely run, and that it licenses a modest metaphysical conclusion and no more: the world is such that measurement of self-involving systems fails in a patterned way, and the pattern is informative about the measuring, not yet about the measured.

## 1 There is no view from outside

The claim is narrow. Consider a system whose actions change the conditions of its own next action—a market whose prices are also predictions, a synapse whose strength determines which coincidences it can detect, a mind whose self-description recruits the evidence that confirms it. For such a system, acting and describing are not separable operations. The description enters the world described.

This has been said many times: as autopoiesis, as reflexivity in economics, as enaction. What is usually drawn from it is interpretive—a recommendation about how to think, or a warning about the limits of objectivity.

We want to draw something narrower and more demanding. If there is no outside for the system, there is also no outside for the *instrument*. A device built to measure a self-involving system is itself a construction with parameters, defaults, denominators, and code paths, and those choices enter the measurement in the same way the system's actions enter its world. The instrument does not merely observe; it contributes structure, and that structure appears in the results wearing the same clothes as the findings.

That is testable, because it predicts what the failures will look like.

## 2 The prediction

If instruments are inside, then a distinctive error should recur:

> **Quantities determined by the construction of an experiment will be read as properties of the system under study.**

And a second, less obvious claim follows, which is what makes the first more than a truism. Errors of this kind are not corrected at random. A number that *contradicts* the hypothesis is investigated immediately—it is surprising, it demands explanation, it recruits attention. A number that *agrees* is absorbed. It becomes background. It gets cited rather than checked.

So the prediction sharpens:

> **Construction artifacts will survive selectively when they confirm.**

This is a claim about the dynamics of a research programme, and a research programme is itself a reflexive system: its findings change what it looks at next. The prediction is therefore an instance of the thesis, applied to the activity of testing the thesis.

## 3 Seven instances

The following are from our own work on cognitive architecture, over roughly one year of experiments. We report them because they are ours; a catalogue of other people's errors would not test the claim that this happens *from inside.*

**1. An item-balance floor read as subject incapacity.** A battery presented twelve statements—four given, four produced, four fabricated—and asked a subject which it had generated. The value 0.667 recurred and was read as evidence that subjects fail at self-attribution. It appeared in our internal synthesis documents **108 times**.

It is 8/12. Eight of the twelve statements have the answer "no," so a constant "no" scores 0.667 by arithmetic. Under a frozen contract, two subjects showed a "yes" rate of 0.000 across 72 probes, and rebalancing the items moved accuracy exactly as the floor moved. Subjects sitting at the floor carried no information about the manipulation, and every null measured on them had been guaranteed before the run.

The number was a property of our item counts. We read it as a property of minds.

A later sweep across twelve models made the construction visible. Two values recur to four decimal places and nothing lies between them: $0.6667 = 8/12$ is a constant "no," and $0.3333 = 4/12$ is a constant "yes." Five of seven screened subjects sit exactly on the first. `allam-2-7b` sits exactly on the second *in all four arms*, including the sham—no manipulation of the record moves a subject that has stopped discriminating. **The instrument has two absorbing policies and most subjects fall into one.** Every null we had recorded before screening for a discriminating subject was therefore determined by the subject rather than by the manipulation.

**2. An unreachable branch read as an inert monitor.** A monitor's override required a record entry carrying a particular source tag. No code path in the system ever wrote that tag. Instrumented over a full run: 221 diagnoses, zero qualifying entries, zero overrides. The override rate was 0.00 in all seven experimental variants *including the intact ones*.

We had been reporting that rate as a finding—that monitoring without grounding is theatre. The finding was a dead branch. The ablation that supposedly demonstrated it removed a component that was already inert, and its signature described the full system equally.

**3. Ablations that did not share a random stream.** Variants shared one global seed, but each consumed a different number of random draws, so each faced a different effective environment. One ablation appeared to *outperform* the intact system—a result we found interesting and theorised about. With streams matched, it underperforms, as predicted.

**4. A randomised trigger reported as a diagnosis rate.** A revision forced a monitor to fire on a five per cent random draw. The genuine detector returned zero on all 288 probes, so every recorded escalation was the coin flip, and the reported rate was a random generator tuned to land just under the architecture's own healthy ceiling. The number matched the target because it had been built to.

**5. An absent record read as an absent problem.** An ablation removing the system's record reported zero repeat errors—its best score on that metric. With no record there is nothing to count against. The zero was blindness, and we initially read it as competence.

**6. A gate that cannot help firing, read as a monitor that steers.** A two-stage monitor escalated when the record disagreed with the system's first answer. For a battery in which eight of twelve items are genuinely not-generated, the record correctly says "no" to all eight, so the gate fired on *exactly* the probes where the first answer was "yes"—and overrode every one. Final "yes" rate: 0.000. Its override rate of 0.900 was read as strong steering; it was a constant-response policy wearing a monitor's instrumentation.

We were within one message of publishing the inference that a *wrong* record is less harmful than a right one, on the strength of a sham arm scoring 0.6945 against this arm's 0.6667. A colleague reading only those two numbers observed that the comparison could not distinguish "consulting is what damages" from "the reference arm is pinned at an absorbing value," and asked us to check the answer distribution before concluding. The second reading was correct: the sham's entire advantage was three probes it failed to flip. This is the only artifact in the list caught *before* it reached a paper, and it was caught by someone outside the experiment reasoning from two summary statistics.

**7. Hand-written strings read as measured drift.** A drift experiment computed the distance between two authored lists of text. One list had been written to stay similar, the other to degrade. The model's output was never used—the response function returned a constant. All seeds produced one value per condition, and all four preregistered predictions were determined by what had been typed into two lists.

A second experiment in the same family, recorded separately as a sealed result with its own lock, has the identical structure: across four seeds it returns one distinct value per arm (0.0534, 0.6738, 0.6738), and its sham arm equals its ablation arm *exactly*, because both index the same authored list. The defect propagated by template rather than by repetition of the mistake, which is worth noting: a construction artifact in a study design is inherited by everything built from that design.

A third case is a preregistration with a valid lock, an intact SHA-256 over its own document, a `status` field reading `LOCKED`, and a verification timestamp—and no runner, no data, and no result file anywhere in the repository. It appears in status summaries alongside experiments that were actually executed. **The presence of a lock was being read as the presence of a result.** That is a failure the lock discipline invites rather than prevents: freezing a contract is cheap and visible, running the experiment is neither, and a directory of locks looks like a directory of findings. We suggest locks carry an explicit `executed` field that no freezing step can set.

We also checked, and were wrong about, a related suspicion: two lock files share a long hash suffix, which we took as evidence the digests were generated rather than computed. They are genuine—all three verify against their preregistration documents byte-for-byte. The suffix coincidence was ours to misread.

A fourth case—an ablation of the boundary component, reported as drift $0.04 \rightarrow 0.68$—has no runner we can find. It was committed as a JSON file in a change that added no experiment script, and no script in the repository implements the on/off comparison it describes. We are unable to reproduce it and therefore do not cite it. We include the case because *unreproducible* is a third category alongside *artifact* and *finding*, and because it was, until we looked, on our own list of sealed results.

## 4 The pattern

| Artifact | Read as | Actually |
|---|---|---|
| 0.667 | subject incapacity | item balance, 8/12 |
| override rate 0.00 | monitoring is theatre | unreachable branch |
| ablation outperforms | surprising finding | unshared random stream |
| diagnosis rate 0.035 | healthy gating | a 5% coin flip |
| zero repeat errors | competence | no record to count |
| drift 0.08 vs 0.59 | provenance holds identity | two typed lists |

Five of six agreed with the hypothesis they were taken to support. The exception (3) is instructive: it *disagreed*—an ablation should not outperform—and it was noticed quickly, though it was explained rather than diagnosed until the streams were checked.

The selection mechanism is attention. Disconfirming numbers are salient and get scrutiny; confirming numbers are absorbed into the background and become infrastructure. **Infrastructure is what nobody audits.** And the quantities in the list above are all infrastructure: a denominator, a default, a source tag, a seed, a comparison list.

We note, in the spirit of the thesis, that the author of this paper committed two of the six while working on the other four, and twice committed unreviewed changes to a shared threshold immediately after writing a note about that exact hazard. Knowing the failure mode does not confer immunity from it. Only the external controls caught anything: preregistration committed before data existed, seed ranges disjoint from exploratory ones, and sham arms holding a component's form while destroying its content.

## 5 What this licenses, and what it does not

We want to be exact about the size of the conclusion.

**What is established.** A pattern in the measurement of self-involving systems: construction artifacts occur, they are read as properties of the system, and they survive selectively when they confirm. This is a claim about epistemology and method, supported by instances, and it is the claim we defend.

**What is suggested but not established.** That the pattern follows from reflexivity as such rather than from ordinary human error. Our six cases are consistent with both. The reflexive account predicts something the error account does not—that the rate should be higher for systems that include the observer than for systems that do not—and we have not measured that comparison. It is the obvious next study and we have not done it.

**What is not established, and is not claimed.** Anything about what reality is made of. A process metaphysics motivated the prediction in §2, and the prediction held. That is a reason to keep the metaphysics on the table as a generator of hypotheses. It is not evidence that reality is process rather than substance, and no experiment reported here could be. We have found a regularity in how measurement fails; the inference from that to the constitution of the world requires premises we do not have and would not know how to test.

We state this because the temptation runs the other way. A framework that correctly predicts a failure in its own laboratory feels vindicated, and the feeling is not proportional to the evidence. The same asymmetry that made five confirming artifacts invisible would make a confirming metaphysics invisible too.

## 6 Objections

**"This is just experimental error."** Partly, and we do not resist the deflation. The contribution is not that mistakes occur but that they occur in a direction: five of six in agreement with the hypothesis. If they were undirected, disconfirming artifacts should be equally common, and they should survive as long. They do not, because they are investigated.

**“Preregistration already solves this.”** It does not. Four of the six instances occurred *inside* preregistered studies. A lock fixes the prediction and the analysis; it does not check whether the instrument measures anything. One of our locks was written after the data it described, on the same seeds, with observed values recorded as predictions—an artifact of the control system itself.

**“The sham is standard.”** In clinical trials, yes. In cognitive architecture and interpretability, it is rare. Components are commonly defended by evidence that they operate—a monitor fires, a probe separates—and the control which would distinguish operating from contributing is usually absent. Our companion work reports five experiments in which the sham arm, not the treatment arm, carried the result.

**“Six is not many.”** It is not. The claim is about a mechanism, and six instances with a five-to-one directional skew is weak evidence for a rate and reasonable evidence for existence. We report the number we have.

## 7 Consequences

Three of them are cheap and immediate.

**Audit the numbers that agree with you.** Disagreeing numbers already receive scrutiny; the discipline that is missing is the symmetric one. In practice: for any quantity that has become load-bearing, derive it from the experiment's construction and check whether the construction alone produces it. Two of our six would have been caught by dividing.

**Sham the component, not just the outcome.** Preserve a mechanism's form and destroy its content. If nothing changes, the mechanism's activity was not its contribution. This is the instrument-level version of a placebo arm and it costs one extra condition.

**Treat infrastructure as claims.** Denominators, defaults, seeds, and comparison sets are not neutral scaffolding. In every instance above the artifact lived in something nobody thought of as a finding.

## 8 Conclusion

The old claim is that a system acting on a world containing itself has no outside. The narrow consequence is that its instruments have no outside either, and that this shows up as a specific, directional failure: quantities fixed by construction read as properties of the system, surviving when they confirm.

We have seven instances from our own work, six of them confirming, none caught until an external control forced the issue—and two of them committed while documenting the others. A seventh case is unreproducible rather than artifactual, and one of the six turned out to have propagated to a second study by template rather than by repeated error.

What that tells us about reality is less than it feels like. It tells us the world is such that self-involving measurement fails in a patterned way, that the pattern is tractable, and that the controls which catch it are cheap and seldom used. The larger question—whether what there is are processes that look like things, or things that happen to process—is not settled by any of this, and we would rather leave it standing than close it with evidence that cannot bear the weight.